

FSY5419/9419,
lecture March 4

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Gates, measurements and
reminder from last week
one-qubit case

$\mathcal{H} = C, Z \leftarrow$ Pauli' z-gate
constant $\in \mathbb{C}$

VQE ansatz $|\psi(\theta)\rangle$

$$\begin{aligned} E[\mathcal{H}] &= \langle E \rangle = \langle \psi(\theta) | \mathcal{H} | \psi(\theta) \rangle \\ &= C, \langle Z \rangle \end{aligned}$$

$$|4(e)\rangle = \alpha|0\rangle + \beta|1\rangle$$

$$\langle E \rangle = C_1 \langle Z \rangle =$$

$$C_1 (\langle 0|\alpha^* + \beta^*\langle 1|) Z (\alpha|0\rangle + \beta|1\rangle)$$

$$\uparrow$$

$$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$= (|\alpha|^2 - |\beta|^2) C_1$$

we perform M -measurements

$$|\alpha|^2 = \frac{m_0}{M} \quad |\beta|^2 = \frac{m_1}{M}$$

$$\langle E \rangle \approx C_1 \left(\frac{m_0}{M} - \frac{m_1}{M} \right)$$

$$Z \otimes I = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix} |00\rangle = + |00\rangle$$



$$\begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} = +1 \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

$|01\rangle$

$$(Z \otimes \mathbb{1}) \begin{matrix} |10\rangle \\ \uparrow \\ \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} \end{matrix} = \underbrace{-1}_{\text{correct when acting with}} |10\rangle$$

$$(Z \otimes \mathbb{1}) |11\rangle = -1 |11\rangle$$

$$1 \otimes Z = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} Z_{2 \times 2} & 0_{2 \times 2} \\ 0_{2 \times 2} & Z_{2 \times 2} \end{bmatrix}$$

$$(1 \otimes Z) |00\rangle = + |00\rangle$$

$$(1 \otimes Z) |10\rangle = + |10\rangle$$

wrong

$$Z \otimes \mathbb{1} = U^\dagger (\mathbb{1} \otimes Z) U$$

$$U = \text{SWAP} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\text{SWAP} |00\rangle = |00\rangle$$

$$\text{SWAP} |01\rangle = |10\rangle$$

$$\text{SWAP} |10\rangle = |01\rangle$$

$$\text{SWAP} |11\rangle = |11\rangle$$

$$Z \otimes I = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} I \otimes Z$$


 Comp basis $\times$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

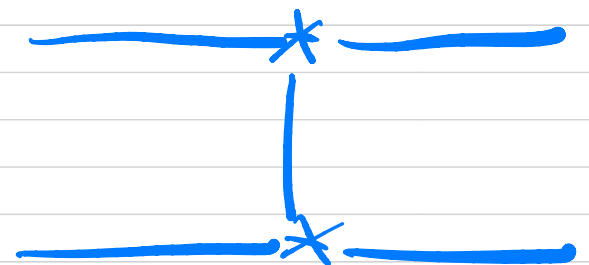
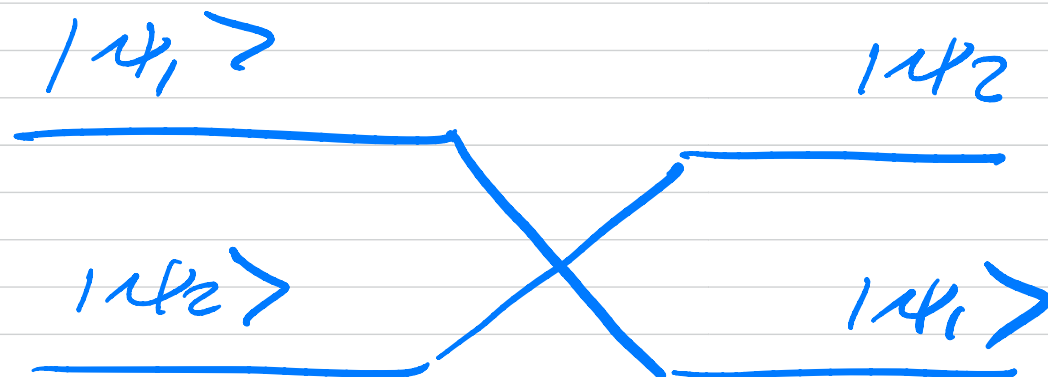
$$\begin{matrix} |0\rangle & \wedge & |1\rangle \\ +1 & & -1 \end{matrix}$$

$$Z |0\rangle$$



$$\begin{aligned}
 X \otimes X &= ? = U_{xx}^\dagger Z \otimes Z U_{xx} \\
 Z \otimes Z &= ? = U_{zz}^\dagger Z \otimes Z U_{zz}
 \end{aligned}$$

$$\text{SWAP} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



$$Z \otimes Z = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$U_{ZZ} = CX10 \quad (\text{two qubit operations})$$

$$U = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

part of gates
called control gates

$$CNOT = CX_{01}$$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} \mathbb{I}_{2 \times 2} & 0_{2 \times 2} \\ 0_{2 \times 2} & X_{2 \times 2} \end{bmatrix}$$

$$CNOT |00\rangle = CNOT \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} = |00\rangle$$

$$CNOT |01\rangle = |01\rangle$$

$$CNOT |10\rangle = |11\rangle$$

$$CNOT |11\rangle = |10\rangle$$

CNOT (CX)

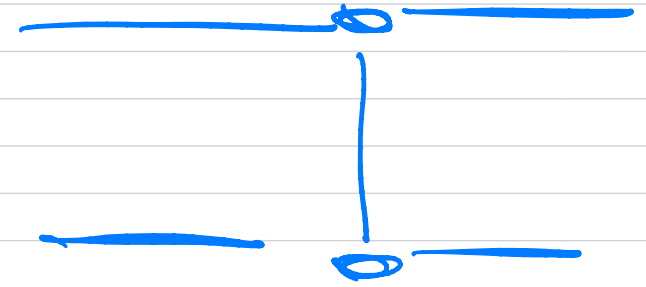
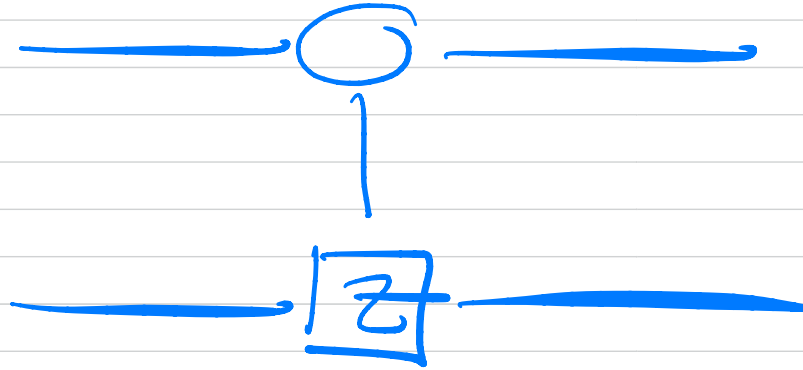
control qubit



if control qubit = $|0\rangle$
nothing happens with
qubit 2 (target)

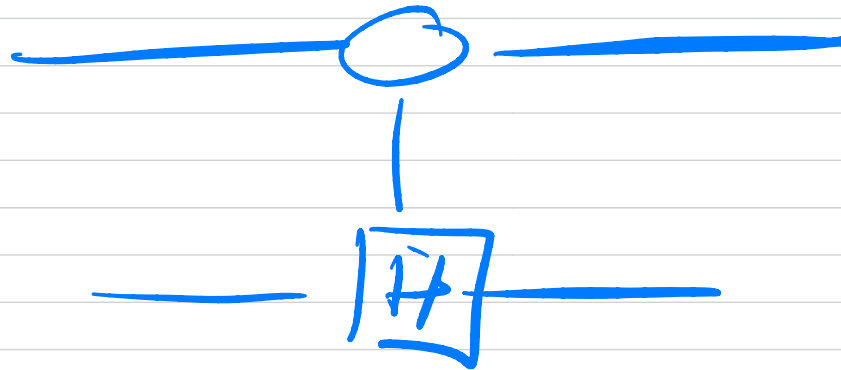
if control equal to $|1\rangle$
qubit 2 changes

Control - z Cz



$Cz |00\rangle = ?$ $Cz |01\rangle = ?$ etc

Control - Hadamard



$$CH = \begin{bmatrix} I_{2 \times 2} & 0_{2 \times 2} \\ 0_{2 \times 2} & H_{2 \times 2} \end{bmatrix}$$

$$CH |00\rangle = |00\rangle$$

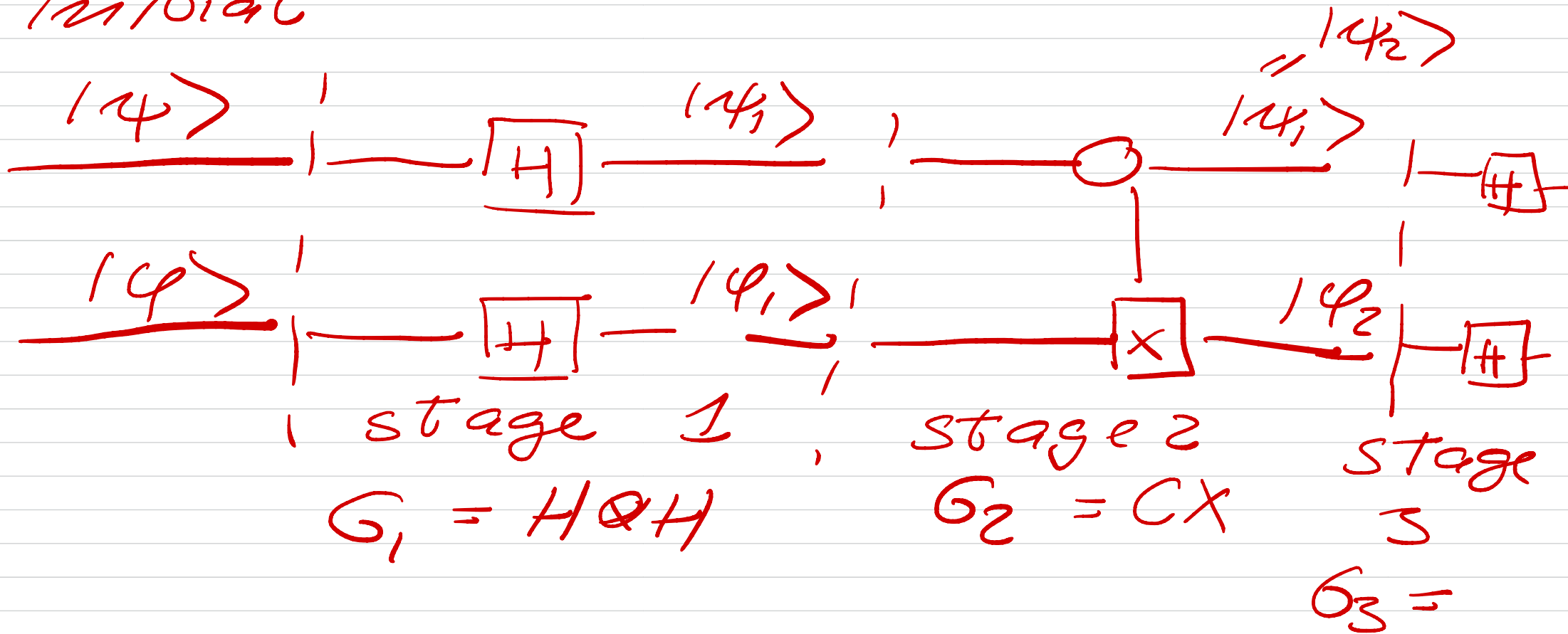
$$CH |01\rangle = |01\rangle$$

$$CH |10\rangle = \frac{1}{\sqrt{2}} [|10\rangle + |11\rangle]$$

$$CH |11\rangle = \frac{1}{\sqrt{2}} [|10\rangle - |11\rangle]$$

Two-qubit system

initial



$$\begin{aligned}
 |4_3 4_3\rangle &= |4_3\rangle \otimes |4_3\rangle \\
 &= G_3 \cdot G_2 \cdot G_1 |44\rangle
 \end{aligned}$$

explicit measurements

$$\langle \psi_1, \psi_2 | Z \otimes I | \psi_1, \psi_2 \rangle$$



final qubit 1 final qubit 2

(depend on
4 angles

$$I \otimes Z$$

SWAP
+

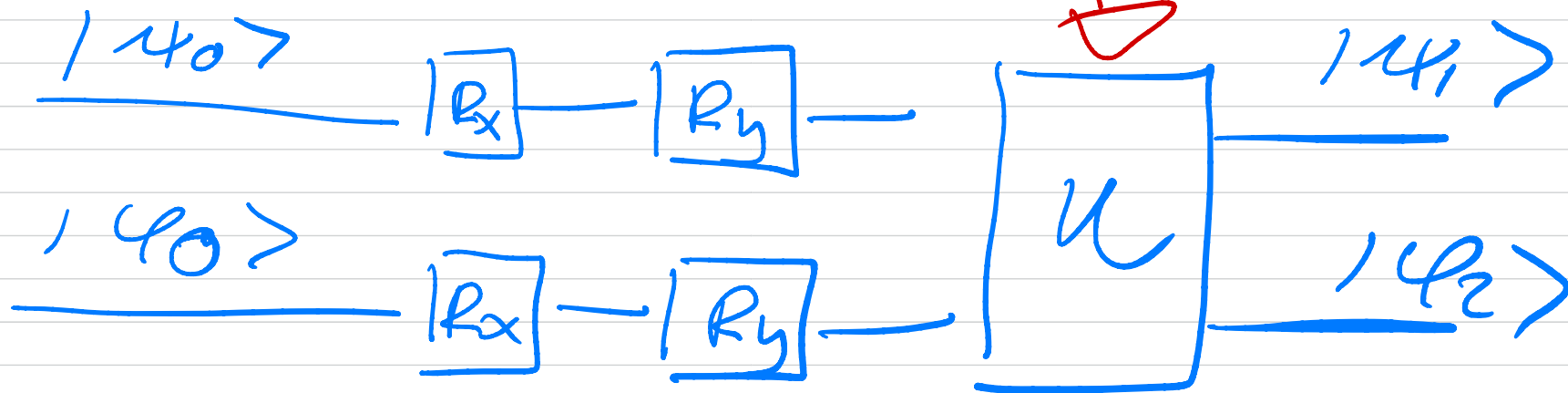
$$\langle \psi_1, \psi_2 | U (Z \otimes I) U | \psi_1, \psi_2 \rangle$$

$X \otimes X$

+

$$\langle \psi_1, \psi_2 | \left(\underbrace{CX_{10} (H \otimes H)} \right) (Z \otimes I) (CX_{10} (H \otimes H))$$

$$\times | \psi_1, \psi_2 \rangle$$



$$z \otimes z$$

+

$$\langle \psi_1, \psi_2 | C X_{10} (z \otimes \mathbb{1}) C X_{10} | \psi_1, \psi_2 \rangle$$

Lipkin model ($J = \frac{1}{2}$)

$$\mathcal{H} = \frac{\varepsilon}{2} \sum_{i=1}^N z_i$$

$$- \frac{V}{2} \sum_{i < j} (x_i x_j - y_i y_j)$$

$$- \frac{W}{2} \sum_{i < j} (x_i x_j + y_i y_j)$$